PHYTOMENADIONE

Newborn prophylaxis at birth

Slow IV — anaphylactoid risk

CHEMICAL NAME 2-methyl-3-[(2E,7R,11R)-3,7,11,15-tetramethylhexadec-2-en-1-yl]naphthalene-1,4-dione (vitamin K1, phylloquinone)

CLASS Fat-soluble vitamin; warfarin antidote

TRADE NAMES Konakion MM · Konakion Neonatal · AquaMephyton · Vitamin K1

THERAPEUTIC USES / INDICATIONS

- Vitamin K deficiency bleeding (haemorrhagic disease) of the newborn — prophylaxis at birth.
- Reversal of warfarin (over-anticoagulation, bleeding).
- Hypoprothrombinaemia from malabsorption, liver disease, prolonged antibiotics, cholestasis.
- Rodenticide (superwarfarin) poisoning.

ADMINISTRATION

- **Neonates:** IM into anterolateral thigh at birth (or oral regimen).
- **IV:** dilute and give slowly over 15–30 min (anaphylactoid reactions with rapid injection).
- **PO:** IV ampoule solution can be given orally (Konakion MM).
- Avoid IM in anticoagulated adults (haematoma).

DOSAGES

- **Newborn:** 1 mg IM at birth (0.5 mg if <2.5 kg/preterm per local policy); oral: 2 mg at birth, day 4–7 and (breastfed) week 4–6.
- **INR 5–8, no bleeding:** stop warfarin 1–2 doses; give 1–5 mg PO if bleeding risk.
- **INR >8, no bleeding:** 1–5 mg PO.
- **Major bleeding:** 5–10 mg IV slowly + prothrombin complex concentrate.

PHARMACOKINETICS

- **Onset:** IV 1–2 h (INR correction 6–12 h); PO 6–10 h. **Duration:** 24–48 h.
- Absorption requires bile salts (PO).
- Metabolised in liver; excreted in bile and urine.

PHARMACODYNAMICS

- Cofactor for **γ-carboxylation** of clotting factors II, VII, IX, X and proteins C and S in liver.
- Overcomes warfarin's blockade of vitamin K recycling.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Injection-site pain, swelling.
- Flushing, taste disturbance (IV).
- Sweating.

ADVERSE EFFECTS (SERIOUS)

- Anaphylactoid reactions with IV injection (cardiac arrest reported).
- Warfarin resistance for days after high doses.
- Haemolysis/kernicterus in neonates (high doses, G6PD; mainly menadione).

CONTRAINDICATIONS & CAUTIONS

- Hypersensitivity.
- Use of menadione (K3) in neonates and G6PD deficiency.

KEY INTERACTIONS

- **Warfarin:** antagonised (high doses → resistance for up to 1–2 weeks).
- **Orlistat, mineral oil, cholestyramine:** ↓ oral absorption.

NURSING MANAGEMENT

- Obtain parental consent for newborn prophylaxis; document dose and route.
- IV: dilute, infuse slowly, monitor for flushing, hypotension, dyspnoea.
- Recheck INR 6–24 h after dose.
- Monitor bleeding signs.

PATIENT EDUCATION

- Explain to parents that newborns are born with low vitamin K; the injection prevents dangerous bleeding.
- Warfarin patients: keep vitamin K intake consistent; follow INR schedule after reversal.

KEY POINTS

- Vitamin K1 — clotting factors II, VII, IX, X.
- Warfarin reversal; IV slowly (anaphylactoid).
- Newborn 1 mg IM at birth.
- INR corrects in 6–12 h; PCC for major bleeds.